Practical No: 5-A

Aim: To develop an Android application using ListView with ArrayAdapter and implement item click handling.

ListView

A ListView is a view component used to display a list of items in a scrollable format.

ArrayAdapter

An ArrayAdapter acts as a bridge between the data source (array) and the ListView.

Item Click Handling

When a user clicks an item in the ListView, an action can be performed using the setOnItemClickListener() method.

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activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical">
  <ListView
     android:id="@+id/listView"
     android:layout_width="match_parent"
     android:layout_height="match_parent"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.adapter_pract;

import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
```

```
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
  ListView listView;

  String[] subjects = {
        "Android",
        "Java",
        "Python",
        "Database",
        "Web Designing",
        "Data Structures",
        "Networking"
  };
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);
    listView = findViewById(R.id.listView);

    ArrayAdapter<String> adapter =
          new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                subjects);

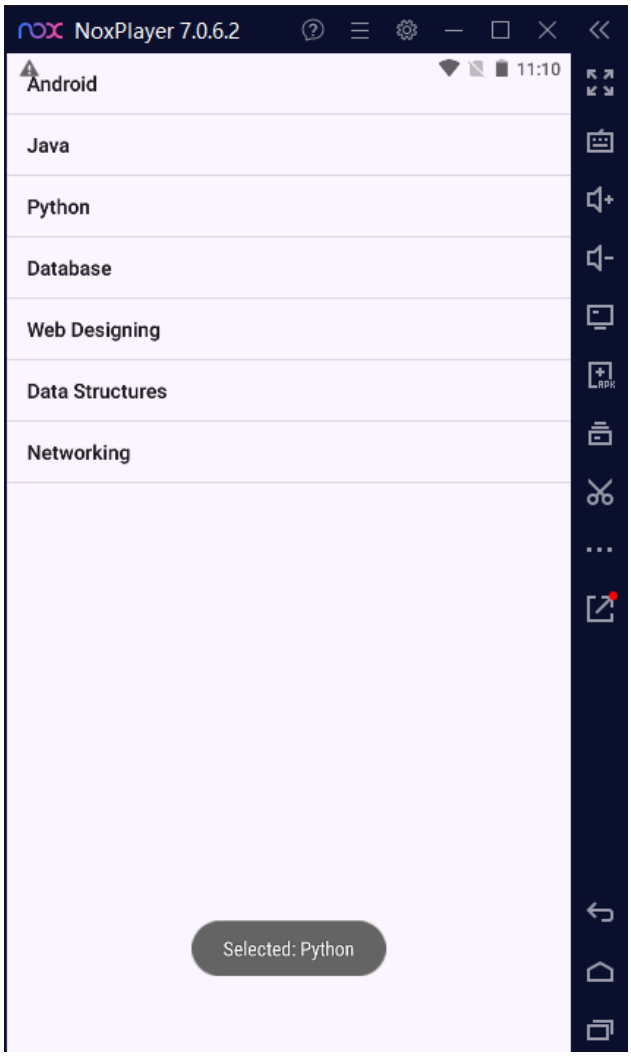
    listView.setAdapter(adapter);

    listView.setOnItemClickListener(
          new AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> parent,
                                    View view,
                                    int position,
                                    long id) {

              Toast.makeText(
                    MainActivity.this,
                    "Selected: " + subjects[position],
                    Toast.LENGTH_SHORT
              ).show();
```

```
            }
        });
    }
}
```

Output:



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